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b)PROBLEM STATEMENT: Create a base class Vehicle with attributes like make, model, and year. Derive a class Car from Vehicle with additional attributes like sittingCapacity,warranty and fuelType. Further, derive a class SportsCar from Car with attributes like topSpeed. Implement member functions to display information about the sports car, including details from all levels of inheritance.

Code:-

```
#include <iostream>
#include <string>
using namespace std;
class Vehicle
{ protected: string make, model; int year;
public: void get_vehicle_data()
 {
 cout << "Enter make and model of the car: " << endl;
 cin >> make >> model;
 cout << "Enter year of manufacture: ";
cin >> year;
 }
 void display_data()
 {
cout << endl;
 cout << "CAR DETAILS" << endl;
cout << "make: " << make << endl; cout << "model: " << model <<
endl;
 cout << "year of manufacture: " << year << endl;
 }
};
class Car : public Vehicle
{ protected:
 int seating_capacity, warranty = 4; string fuel_type;
public: void get_car_data()
 {
 get_vehicle_data();
 cout << "Enter seating capacity: ";
cin >> seating_capacity; cout << "Enter fuel type of car: ";
cin >> fuel_type;
 }
 void display_car_data()
 {
 display_data();
 cout << "Seating capacity: " << seating_capacity << endl;
cout << "Warranty provided is " << warranty << " years" << endl;
cout << "Fuel type: " << fuel_type << endl;
 }
};
class Sports_Car : public Car
{ private: int top_speed;
public: void get_Sports_Car_data()
```

```
 {
 get_car_data();
 cout << "Enter top speed of the car: ";
cin >> top_speed;
 }
 void display_Sports_Car_data()
 {
 display_car_data();
 cout << "Top speed: " << top_speed << "km/hr" << endl;
 }
}; int main()
{
 Sports_Car s;
 s.get_Sports_Car_data();
 s.display_Sports_Car_data();
 return 0;
}
```

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OUTPUT:-

```
Enter make and model of the car:
Mercedes
G35
Enter year of manufacture: 2022
Enter seating capacity: 2
Enter fuel type of car: Petrol
Enter top speed of the car: 300

CAR DETAILS
make: Mercedes
model: G35
year of manufacture: 2022
Seating capacity: 2
Warranty provided is 4 years
Fuel type: Petrol
Top speed: 300km/hr
```